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WORK VEHICLE PATH PLAN GENERATION SYSTEM AND WORK VEHICLE PATH PLAN GENERATION METHOD

Abstract

An aspect of the present disclosure provides a work vehicle path plan generation system that generates a path plan in order for a work vehicle including work equipment to perform an excavation work on a ground of a construction target area. The work vehicle path plan generation system includes a position detection unit configured to detect a position of the work vehicle, an information storage unit configured to store topographic shape information indicating a shape of a topography in the construction target area, a position of the work vehicle, and design surface information indicating a shape required to be excavated of the ground in the construction target area, and a path plan generation unit configured to generate a work equipment path plan indicating a movement path of the work equipment and a travel path plan indicating a travel path of the work vehicle, based on the topographic shape information, the position of the work vehicle, and the design surface information.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a work vehicle path plan generation system and a work vehicle path plan generation method.

The present application claims priority with respect to Japanese Patent Application No. 2022-106372 filed in Japan on Jun. 30, 2022, the contents of which are incorporated herein by reference.

BACKGROUND ART

[0002] In an excavation plan creation device disclosed in Patent Document 1, a machine learning model is used in which topographic information is input and planned values of an excavation trajectory and a swing direction are output. This excavation plan creation device generates a plurality of plan models subjected to machine learning by changing each parameter related to soil quality with an excavation efficiency as an evaluation value. Further, this excavation plan creation device estimates the soil quality, selects the plan model based on the estimated soil quality, inputs the topographic information to the selected plan model, and calculates the planned value as an output of the plan model. In addition, this excavation plan creation device estimates the topographic information based on time-series data of a teeth position of a bucket and a load of work equipment that supports the bucket, and the estimated soil quality.

CITATION LIST

Patent Document

[Patent Document 1]

[0003] Japanese Unexamined Patent Application, First Publication No. 2021-188362

SUMMARY OF INVENTION

Technical Problem

[0004] In the excavation plan creation device disclosed in Patent Document 1, it is possible to create an excavation plan with high accuracy in consideration of the influence of the soil quality appropriately in a case where the excavation plan is created. On the other hand, for example, it is necessary to perform estimation calculation or the like based on the time-series data of the teeth position of the bucket and the load of the work equipment that supports the bucket, and the soil quality. In particular, in a case where an earth is handled as a particle model for the soil quality, there is a possibility that a load of calculation processing may be increased.

[0005] The present disclosure has been made in view of the above circumstances, and an object of the present disclosure is to provide a work vehicle path plan generation system and a work vehicle path plan generation method capable of efficiently generating a path plan for the work vehicle.

Solution to Problem

[0006] In order to achieve the aforementioned objects, one aspect of the present disclosure provides a work vehicle path plan generation system that generates a path plan in order for a work vehicle including work equipment to perform an excavation work on a ground of a construction target area. The work vehicle path plan generation system includes a position detection unit configured to detect a position of the work vehicle, an information storage unit configured to store topographic

shape information indicating a shape of a topography in the construction target area, a position of the work vehicle, and design surface information indicating a target shape in the construction target area, and a path plan generation unit configured to generate a work equipment path plan indicating a movement path of the work equipment and a travel path plan indicating a travel path of the work vehicle, based on the topographic shape information, the position of the work vehicle, and the design surface information.

Advantageous Effects of Invention

[0007] According to each aspect of the present disclosure, it is possible to efficiently generate the path plan of the work vehicle.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 A plan view of a construction target area in which an excavation work is performed by a work vehicle according to an embodiment.

[0009] FIG. 2 A side-sectional view of the construction target area in which the excavation work is performed by the work vehicle according to an embodiment.

[0010] FIG. 3 A schematic block diagram showing a configuration example of a path plan generation device according to an embodiment.

[0011] FIG. 4 A diagram showing an example of a movement path of work equipment in a travel path of the work vehicle according to an embodiment.

[0012] FIG. 5 A diagram showing an example of a simulation model used for a simulation for candidates for the movement path in a path plan generation unit according to an embodiment.

[0013] FIG. 6 A diagram showing another example of the movement path of the work equipment in the travel path of the work vehicle according to an embodiment.

[0014] FIG. 7 A diagram showing still another example of the movement path of the work equipment in the travel path of the work vehicle according to an embodiment.

[0015] FIG. 8 A side view of an earth held in front of a blade and an earth of windrows formed by overflowing onto both sides of the blade, in a case where excavation is performed by the work equipment according to an embodiment.

[0016] FIG. 9 A plan view of the earth held in front of the blade and the earth of the windrows formed by overflowing onto both sides of the blade, in a case where the excavation is performed by the work equipment according to an embodiment.

[0017] FIG. 10 A diagram in which the earth held in front of the blade and the earth of the windrows formed by overflowing onto both sides of the blade, in a case where the excavation is performed by the work equipment according to an embodiment, are modeled as a polygonal prism-shaped three-dimensional model.

[0018] FIG. 11 A flowchart showing an operation of the path plan generation device according to an embodiment.

[0019] FIG. 12 A diagram showing an outline of a flow of generating a path plan with learning by reinforcement learning according to an embodiment.

[0020] FIG. 13 A schematic block diagram showing a configuration example of a work vehicle path plan generation system according to an embodiment.

DESCRIPTION OF EMBODIMENTS

[0021] Hereinafter, embodiments of the present disclosure will be described with reference to drawings. In each drawing, the same reference numerals are used for the same or corresponding configurations, and the description thereof will be omitted as appropriate.

(Configuration Example of Work Vehicle 100)

[0022] FIG. 1 is a plan view of a construction target area A in which an excavation work is

performed by a work vehicle **100** (work machine) according to the embodiment. FIG. 2 is a side-sectional view of the construction target area A in which the excavation work is performed by the work vehicle **100** according to the embodiment.

[0023] As shown in FIGS. 1 and 2, in the present embodiment, the work vehicle **100** performs the excavation work on a ground G of the predetermined construction target area A. The work vehicle **100** excavates the ground G to form an excavated ground surface K along a design surface S designed in advance. The construction target area A shown in FIGS. 1 and 2 is merely an example, and a planar shape and the like thereof can be changed as appropriate. In addition, a topographic shape of a ground surface of the ground G and a shape of the design surface S (excavated ground surface K) are merely examples and can be changed as appropriate.

[0024] The work vehicle **100** is automatically operated by a remote operation at a construction site including the construction target area A to excavate the ground G. The work vehicle **100** according to the embodiment is a bulldozer as an example. The work vehicle **100** includes an undercarriage **110**, an upper vehicle body **120**, and work equipment **130**.

[0025] The undercarriage **110** supports the work vehicle **100** in a travelable manner. The undercarriage **110** includes, for example, a pair of left and right crawlers of a crawler **110a** (also referred to as left crawler **110a**) and a crawler **110b** (also referred to as right crawler **110b**). Both left and right crawlers **110a** and **110b** can be driven independently drive wheels to move forward and move rearward. In a case where the left crawler **110a** and the right crawler **110b** are moved forward at the same time, the undercarriage **110** moves forward. In a case where the left crawler **110a** and the right crawler **110b** are moved rearward at the same time, the undercarriage **110** moves rearward. In addition, in a case where the drive wheel of one crawler and the drive wheel of the other crawler are driven in opposite directions to each other, for example, in a case where the right crawler **110b** is moved forward and the left crawler **110a** is moved rearward at the same time, the undercarriage **110** can rotate around a swing center.

[0026] The upper vehicle body **120** is supported on the undercarriage **110**. The upper vehicle body **120** includes a cab **121**. The cab **121** is a space where an operator (driver) boards to operate the work vehicle **100**.

[0027] The work equipment **130** includes at least a lift frame **131** and a blade **133**. The lift frame **131** is attached to the undercarriage **110** in an operable manner. The blade **133** excavates an earth or the like. The blade **133** is attached to the lift frame **131** in an operable manner.

[0028] As shown in FIG. 1, the work vehicle **100** excavates the ground G with the blade **133** while moving along a plurality of travel paths R in the construction target area A. In FIG. 1, each of the plurality of travel paths R is linear in a plan view. However, the travel paths R are not limited to being linear, and may be curved or bent as appropriate in accordance with topography, an obstacle, or the like. In addition, in FIG. 1, the plurality of travel paths R are set to be parallel to each other in a plan view, but may be, for example, extended in a radial shape.

(Configuration Example of Path Plan Generation Device **20**)

[0029] FIG. 3 is a schematic block diagram showing a configuration example of a path plan generation device **20** according to the embodiment.

[0030] As shown in FIG. 3, the path plan generation device **20** can be configured by using a computer, such as a microcomputer or a central processing unit (CPU), and hardware, such as a peripheral circuit of the computer or a peripheral device. The path plan generation device **20** includes an information input unit **21**, an information storage unit **22**, a path plan generation unit **23**, and an information output unit **24**, as a functional configuration configured of a combination of the hardware and software such as a program executed by the computer.

[0031] The information input unit **21** receives, from the outside, inputs of topographic shape information indicating the topographic shape of the ground surface or the like of the ground G in the construction target area A, a position of the work vehicle **100**, and design surface information indicating a target shape in the construction target area A. The design surface information is

acquired from, for example, an external computer aided design (CAD) system that designs the construction site including the construction target area A. The topographic shape information is acquired by a detection device, such as a radar, provided in the work vehicle **100**. The detection device may be provided in another work vehicle, or may be attached to a structure in the construction area A. In addition, the detection device may be mounted on a flying object that flies over the construction site. An example of the flying object includes an unmanned aerial vehicle (UAV) such as a drone.

[0032] The information storage unit **22** stores the topographic shape information and the design surface information, which are received through the information input unit **21**. In addition, the information storage unit **22** stores various types of vehicle information related to the work vehicle **100**, for example, the size of the blade **133**, the travel driving force by the crawlers **110a** and **110b**, and the travel maximum speed.

[0033] As will be described in detail below, the path plan generation unit **23** generates a path plan for performing an excavation construction in the construction target area A by the work vehicle **100**.

[0034] The information output unit **24** outputs information about the path plan generated by the path plan generation unit **23** to the outside.

(Configuration Example of Path Plan Generation Unit **23**)

[0035] FIG. **4** is a diagram showing an example of a movement path L of the work equipment **130** in the travel path R of the work vehicle **100** according to the embodiment.

[0036] The path plan generation unit **23** generates a work equipment path plan and a travel path plan of the work vehicle **100**, as path plans for performing the excavation construction in the construction target area A by the work vehicle **100**. The movement path L of the work equipment **130** indicates, in a case where the work vehicle **100** travels along the travel path R, positions and angles of the blade **133** in a vertical direction at a plurality of positions on the travel path R.

[0037] The path plan generation unit **23** generates the work equipment path plan that is an optimal movement path L of the work equipment **130**, and the travel path plan that is an optimal travel path R of the work vehicle **100**. The path plan generation unit **23** generates the work equipment path plan and the travel path plan in which slip of the crawlers **110a** and **110b** of the work vehicle **100** does not occur and the excavation efficiency by the blade **133** is improved. In the path plan generation unit **23**, reinforcement learning is performed to generate the work equipment path plan that is the optimal movement path L of the work equipment **130**, and the travel path plan that is the optimal travel path R of the optimal work vehicle **100**, for a plurality of candidates for the movement path L and the travel path R. In the present embodiment, a plurality of times of simulations for the plurality of candidates for the movement path L and the travel path R in which parameters related to an operation of the work equipment **130** are variously changed are performed while performing learning by the reinforcement learning.

[0038] FIG. **5** is a diagram showing an example of a simulation model used for a simulation for candidates for the movement path L and the travel path R in the path plan generation unit **23** according to the embodiment.

[0039] The path plan generation unit **23** decomposes a phenomenon that occurs during the excavation by the work vehicle **100** into a model for each element, in order to perform the simulation on the candidates for the movement path L and the travel path R. For example, as shown in FIG. **5**, in the path plan generation unit **23**, the phenomena occurring during the excavation by the work vehicle **100** is decomposed into a vehicle body model **M10**, a control model **M20**, and an earth model **M30**, and the reinforcement learning is performed by the simulation. The simulation is executed by a simulator that is operated on a computer to execute a simulation.

[0040] In the vehicle body model **M10**, parameters related to the operation of the work vehicle **100** on the travel path R and parameters related to the operation of the work equipment **130** on the travel path R are related. More specifically, the vehicle body model **M10** includes, for example, a

hydraulic pressure model **M11** related to hydraulic equipment, such as a lift cylinder and a tilt cylinder, of the work equipment **130**, a mechanism model **M12** related to a mechanism of a movable part, such as the lift frame **131** and the blade **133**, and a vehicle-underbody model **M13** related to the undercarriage **110**.

[0041] In the hydraulic pressure model **M11**, for example, a relief pressure or the like related to the hydraulic equipment, such as the lift cylinder and the tilt cylinder, of the work equipment **130** is simulated. In such hydraulic equipment, a maximum value of a reaction force caused by the operation of the lift frame **131** and the blade **133** is set as a maximum reaction force restriction. In the hydraulic pressure model **M11**, the relief pressure or the like of the hydraulic equipment, which is caused by the operation of the work equipment **130**, is simulated within a range not exceeding the maximum reaction force restriction.

[0042] In the mechanism model **M12**, the simulation is performed for an excavation range in which the blade **133** can excavate in a case where the movable part, such as the lift frame **131** and the blade **133**, is operated based on the movement path **L**. A position restriction of a tip blade is set in the blade **133**. In the mechanism model **M12**, the simulation of the excavation operation by the blade **133** is performed within a range not exceeding the position restriction of the tip blade.

[0043] In the vehicle-underbody model **M13**, the simulation is performed for a vehicle speed due to the drive of the undercarriage **110**, a degree of occurrence of the slip of the crawlers **110a** and **110b**, and the like. The vehicle speed in a case where the work vehicle **100** travels and moves along the travel path **R** by the drive of the undercarriage **110** is calculated based on the travel driving force (traction force) of the crawlers **110a** and **110b** and a reaction force received by pushing the earth of the ground **G** with the blade **133**. For example, the degree of occurrence of the slip of the crawlers **110a** and **110b** can be simulated based on the reaction force received from the earth in a case where the lift frame **131** and the blade **133** perform the excavation operation. This reaction force is a sum of an excavation resistance (shear resistance) in a case where the blade **133** excavates the ground **G** and an earth moving resistance due to friction in a case where the earth held in front of the blade **133** is pushed to the front. In the vehicle-underbody model **M13**, in a case where the reaction force reaches the maximum reaction force restriction, determination can be made that a shoe slip limit of the crawlers **110a** and **110b** is exceeded and the slip occurs.

[0044] The control model **M20** relates to a control condition in a case where the simulation is performed based on the candidates for the movement path **L** and the travel path **R**. The control model **M20** has, for example, a path tracking model **M21** and a path plan model **M22**.

[0045] In the path tracking model **M21**, it is assumed that, in a case where the simulation is performed, trajectory trackability to the travel path **R** of the work vehicle **100** and trajectory trackability to the movement path **L** of the work equipment **130** each have trackability of 100%, for example.

[0046] In addition, in the path plan model **M22**, the path plan model **M22** for the plurality of movement paths **L** and the travel paths **R**, which are learning targets in performing the plurality of times of simulations, is set. In the path plan model **M22**, new movement paths **L** and travel paths **R** are generated such that the excavation efficiency is further improved, based on results of the simulations performed for the candidates for the movement path **L** and the travel path **R**. As shown in FIG. 4, in the path plan model **M22**, for example, a plurality of candidates for the movement path **L** to be simulated are sequentially generated by variously changing conditions, such as a start point **P1** of the movement path **L**, an excavation start position **P2** at which the blade **133** is lowered to start the excavation of the ground **G**, an excavation depth **P3** at which the ground **G** is excavated by one excavation, an entry angle **P4** of the blade **133** with respect to the ground **G**, an end point **P5** of the movement path **L**, and the vehicle speed of the work vehicle **100**, which are provisionally set along the travel path **R**. In the path plan model **M22**, the simulation is performed for one candidate for the movement path **L**, then another movement path **L** having a different condition is generated, and the simulation is sequentially executed.

[0047] The earth model **M30** is for simulating parameters related to the earth of the ground **G** to be excavated by the work equipment **130** in a case where the work vehicle **100** is caused to travel based on each candidate of the movement path **L** and the travel path **R** to move the work equipment **130**. The earth model **M30** is classified into a topographic model **M31** related to a topographic change caused by the excavation of the ground **G** with the blade **133** and a reaction force model **M32** related to the reaction force received by the blade **133** in a case where the ground **G** is excavated.

[0048] In the topographic model **M31**, the simulation is performed for an excavation earth amount to be excavated by the excavation of the ground **G** with the blade **133**. The excavation earth amount is calculated based on a difference between the topographic shape information of the ground **G** before the excavation and the design surface information for the design surface **S**, in a region along the travel path **R**. In the topographic model **M31**, in a case where the excavation from the ground **G** to the design surface **S** is divided into a plurality of times, the excavation earth amount at each time of the excavation can be calculated based on a difference between the topographic shape before the excavation and an excavation surface formed by one excavation along the movement path **L**.

[0049] FIG. **6** is a diagram showing another example of the movement path **L** of the work equipment **130** in the travel path **R** of the work vehicle **100** according to the embodiment.

[0050] In addition, as shown in FIG. **6**, in a case where the ground **G** is excavated by the blade **133** based on the movement path **L** and an earth **D1** is held in front of the blade **133**, the simulation is performed for a holding earth amount, which is the amount of the held earth **D1**, in the topographic model **M31**.

[0051] FIG. **7** is a diagram showing still another example of the movement path **L** of the work equipment **130** in the travel path **R** of the work vehicle **100** according to the embodiment.

[0052] As shown in FIG. **7**, in a case where the other part of the ground **G** on the travel path **R** is backfilled by an earth **D2** obtained by the excavation of a part of the ground **G** on the travel path **R** with the blade **133**, the simulation may be performed for an actual excavation earth amount based on a difference between the excavation earth amount and a backfill earth amount, which is an amount of the backfilled earth **D2**, in the topographic model **M31**.

[0053] FIG. **8** is a side view of the earth **D1** held in front of the blade **133** and an earth **D5** of windrows formed by overflowing onto both sides of the blade **133** in a case where the excavation is performed by the work equipment **130** according to the embodiment. FIG. **9** is a plan view of the earth **D1** held in front of the blade **133** and the earth **D5** of the windrows formed by overflowing onto both sides of the blade **133** in a case where the excavation is performed by the work equipment **130** according to the embodiment.

[0054] In the topographic model **M31**, as shown in FIGS. **8** and **9**, in a case where the ground **G** is excavated by the blade **133** based on the movement path **L** and the travel path **R**, the simulation is performed for an amount of the earth **D5** of the so-called windrows that overflow onto both sides of the blade **133** in a width direction. In the topographic model **M31**, in a case where the ground **G** is excavated by the blade **133** based on the movement path **L** along one travel path **R** to form the windrow on the ground **G**, the amount of the earth **D5** of the formed windrow is included in the topographic information in a case where the simulation is performed for the candidate for the movement path **L** in another travel path **R**.

[0055] The amount of the held earth **D1** and the amount of the earth **D5** of the windrow may be calculated, for example, based on a preliminary experiment using a model.

[0056] In addition to the above, in the topographic model **M31**, for example, in a case where a part of the earth excavated on the travel path **R** is backfilled at another position on the travel path **R**, the simulation is performed by assuming that the backfilled earth is rolled with the crawlers **110a** and **110b** for compaction.

In the topographic model **M31**, the simulation is also performed for a landslide for the earth excavated on the travel path **R**.

[0057] In the reaction force model M32, in a case where the blade 133 is operated while causing the work vehicle 100 to move along the travel path R to excavate the ground G based on the movement path L, the simulation is performed for the earth moving resistance that occurs between the blade 133 and the ground G in a case where the earth D1 held in front of the blade 133 is pushed to the front. In a case where the work vehicle 100 moves forward along the travel path R, the amount of the earth D1 held in front of the blade 133 gradually increases. That is, the earth moving resistance changes every moment.

[0058] In the reaction force model M32, in a case where the blade 133 is operated along the movement path L, the simulation is performed for the excavation resistance received by the blade 133 from the earth of the ground G. The excavation resistance increases with the excavation depth by the blade 133 into the ground G. As shown in FIG. 7, in a case where the earth D2 is backfilled in front of the end point P5 of the movement path L, the reaction force model M32 also simulates the excavation resistance received by the blade 133 from the backfilled earth D2.

[0059] The simulator performs the reinforcement learning on the path plan generation unit 23 based on the results of the plurality of times of simulations performed for the plurality of candidates for the movement path L and the travel path R. For each simulation, based on the excavation earth amount in a case where the work equipment 130 is operated along the movement path L, while causing the work vehicle 100 to move along the travel path R, and a work time required for the work equipment 130 to perform the excavation from the start point to the end point of the movement path L, the simulator calculates a reward by the reinforcement learning, for example, based on the following equation (1).

[00001] $\text{Reward} = \text{excavationearthamount} / \text{worktime} \quad (1)$

[0060] In addition, for each simulation, in a case where the work equipment 130 is operated along the movement path L while causing the work vehicle 100 to move along the travel path R, the simulator provides a penalty value (for example, -0.05) in the reinforcement learning in a case where the reaction force received by the blade 133 from the earth exceeds the shoe slip limit of the undercarriage 110.

[0061] With the reinforcement learning, the simulator searches for optimal movement path L and travel path R, based on the calculated reward and the penalty value, and the path plan generation unit 23 learns to generate the work equipment path plan and the travel path plan.

[0062] The path plan generation unit 23 is learned to generate the work equipment path plan and the travel path plan for the entire construction target area A based on the work equipment path plan, which is the optimum movement path L, and the travel path plan, which is the optimum travel path R.

[0063] FIG. 10 is a diagram in which the earth D1 held in front of the blade 133 and the earth D5 of the windrows formed by overflowing onto both sides of the blade 133, in a case where the excavation is performed by the work equipment 130 according to the embodiment, are modeled as polygonal-shaped three-dimensional models Dm1 and Dm5.

[0064] Meanwhile, as shown in FIGS. 8 to 10, in the topographic model M31 of the path plan generation unit 23 as described above, the earth D1 held in front of the blade 133 is simplified by modeling the earth D1 as the polygonal prism-shaped (for example, triangular prism-shaped) three-dimensional model Dm1 to calculate the holding earth amount. Accordingly, the holding earth amount that changes every moment in accordance with the excavation of the ground G based on the movement path L can be efficiently calculated.

[0065] In addition, in the topographic model M31, the amount of the earth D5 of the windrows that protrude on both sides of the blade 133 is calculated in a simplified manner by modeling the earth D5 as the polygonal prism-shaped (for example, quadrangular prism-shaped (rectangular prism)) three-dimensional model Dm5 in accordance with a ratio of the earth D1 held in the front.

Accordingly, the amount of the earth D5 of the windrows that changes every moment in accordance

with the excavation of the ground G based on the movement path L can be efficiently calculated.
(Operation Example of Path Plan Generation Device 20)

[0066] FIG. 11 is a flowchart showing an operation of the path plan generation device 20 according to the embodiment.

[0067] The path plan generation device 20 learned by the reinforcement learning is provided in, for example, the work vehicle 100. The path plan generation device 20 generates the path plan for performing the excavation work on the ground G of the construction target area A by the work vehicle 100.

[0068] For this purpose, as shown in FIG. 11, first, the information input unit 21 acquires, from the outside, the topographic shape information indicating the shape of the ground G in the construction target area A, the position of the work vehicle 100, and the design surface information indicating a shape required to be excavated of the ground G in the construction target area A (S1). Next, the information storage unit 22 stores the acquired topographic shape information, position of the work vehicle 100, and design surface information (S2). Next, the path plan generation unit 23 generates the work equipment path plan indicating the movement path L of the work equipment 130 and the travel path plan indicating the travel path R of the work vehicle 100, based on the topographic shape information, the position of the work vehicle 100, and the design surface information (S3). Thereafter, the information output unit 24 outputs the generated work equipment path plan and travel path plan to the outside (S4).

[0069] FIG. 12 is a diagram showing an outline of a flow in which the path plan generation unit 23 is learned through the reinforcement learning according to the embodiment. The reinforcement learning can be performed, for example, by using the simulator that operates on a computer outside the vehicle body 100 to execute the simulation.

[0070] For this purpose, first, the simulator sets the candidate for the movement path L of the work equipment 130 and the candidate for the travel path R of the work vehicle 100 (S31).

[0071] The simulator performs the simulation based on the set candidates for the movement path L and the travel path R of the work vehicle 100. Specifically, the simulation for the operations of the work vehicle 100 and the work equipment 130 is performed using the vehicle body model M10 and the control model M20 (S32). In the simulation of the operation of the work vehicle 100, a travel trajectory of the work vehicle 100 along the travel path R in a case where the work equipment 130 is operated is simulated based on the set candidates for the movement path L and the travel path R of the work vehicle 100. In addition, in the simulation of the operation of the work equipment 130, a movement trajectory of the work equipment 130 in a case where the work equipment 130 is operated based on the movement path L is simulated.

[0072] In addition, based on data of the calculated travel trajectory of the work vehicle 100 and movement trajectory of the work equipment 130, the simulator calculates the work time in a case where the excavation is performed based on the movement path L and the travel path R.

[0073] Further, based on the data of the travel trajectory of the work vehicle 100 and the movement trajectory of the work equipment 130, the simulator executes the simulation for the earth in a case where the earth of the ground G is excavated, by using the earth model M30 (S33). In the earth model M30, for example, the excavation resistance in a case where the earth of the ground G is excavated is calculated. Data of the calculated excavation resistance is fed back to the vehicle body model M10 and is reflected in the calculation of the vehicle speed or the like of the work vehicle 100.

[0074] In addition, the simulator uses the earth model M30 to calculate a post-excavation topography indicating the shape of the ground G after the excavation by the work equipment 130. Further, the path plan generation unit 23 calculates the excavation earth amount by the work equipment 130, the holding earth amount, the earth amount of the windrow, and the like, using the earth model M30.

[0075] The simulator learns the path plan generation unit 23 by the reinforcement learning based

on the work time and the excavation earth amount calculated as described above (S34). In the reinforcement learning, the reward and the penalty are calculated to evaluate the candidates for the movement path L and the travel path R. In this case, the simulator determines whether or not the excavation resistance exceeds the shoe slip limit of the crawlers **110a** and **110b** in a case where the work equipment **130** performs the excavation based on the movement path L while causing the work vehicle **100** to move along the travel path R. As a result, in a case where the excavation resistance exceeds the shoe slip limit, determination is made that the slip of the work vehicle **100** occurs.

[0076] In the simulator, after the simulation for one candidate for the movement path L and the travel path R is performed, the learning based on an evaluation result for the candidate for the movement path L and the travel path R is performed, and the candidate for the movement path L or the travel path R required to be simulated next is set. The candidate for the movement path L or the travel path R required to be simulated next is set such that the slip of the work vehicle **100** does not occur and the reward represented by the above equation (1) is as large as possible.

[0077] The simulation for the candidate for the movement path L and the travel path R is repeated a plurality of times while performing the learning by the reinforcement learning. Accordingly, the path plan generation unit **23** is learned such that the slip of the work vehicle **100** does not occur and the work equipment path plan and the travel path plan having a high excavation efficiency can be generated.

(Configuration of Work Vehicle Path Plan Generation System)

[0078] FIG. **13** is a schematic block diagram showing a configuration example of a work vehicle path plan generation system **50** according to the embodiment.

[0079] As shown in FIG. **13**, the work vehicle path plan generation system **50** includes a remote control device **60** and the work vehicle **100**.

[0080] The remote control device **60** includes a communication unit **61**, an information output unit **62**, and a supervision unit **63**.

[0081] The communication unit **61** can communicate with the work vehicle **100** by a public wireless communication network and a wireless communication unit.

[0082] The information output unit **62** outputs information necessary for automatically driving the work vehicle **100**. The information output unit **62** acquires, for example, the design surface information or the like of the construction target area A from an external CAD system or the like, and transmits the acquired information to the work vehicle **100** via the communication unit **61**.

[0083] The supervision unit **63** monitors an operation state of each unit of the work vehicle **100** based on information detected by various sensors provided in the work vehicle **100**.

[0084] The work vehicle **100** is automatically operated based on the path plan generated by the path plan generation device **20**.

[0085] Therefore, the work vehicle **100** includes a communication unit **71**, the path plan generation device **20**, a position detection unit **72**, a path plan storage unit **73**, and a vehicle control unit **74**.

[0086] The communication unit **71** can communicate with the communication unit **61** of the remote-control device **60** by a public wireless communication network and a wireless communication unit.

[0087] A part or all of the configuration of the work vehicle **100** may be provided in the remote control device **60**. For example, the path plan generation device **20** described above is provided on the work vehicle **100** side in the present embodiment. The path plan generation device **20** may be provided on the remote control device **60** side. In addition, a part or all of the configuration of the remote control device **60** may be provided in the work vehicle **100**.

[0088] The position detection unit **72** is provided in the work vehicle **100**. The position detection unit **72** can detect the position of the work vehicle **100** using, for example, a GPS or the like.

[0089] The path plan storage unit **73** stores the work equipment path plan and the travel path plan of the work vehicle **100**, which are generated by the path plan generation device **20** and are output

to the outside.

[0090] The vehicle control unit **74** controls the operation of each unit of the work equipment **130** and the work vehicle **100**, based on the work equipment path plan and the travel path plan stored in the path plan storage unit **73**. The work vehicle **100** operates the work equipment **130** while causing the work vehicle **100** to move based on the work equipment path plan, which is the optimum movement path L of the work equipment **130**, and the travel path plan, which is the optimum travel path R of the work vehicle **100**. Accordingly, the excavation of the ground G is efficiently performed.

Actions and Effects

[0091] According to the present embodiment, the path plan of the work vehicle **10** can be efficiently generated.

[0092] Although the embodiments of the present disclosure have been described with reference to the drawings, the specific configuration is not limited to the above-described embodiments, and the design change and the like within a range not departing from the scope of the present disclosure are also included.

[0093] For example, the work vehicle **100** according to the embodiments described above is a bulldozer, but the present disclosure is not limited thereto. For example, the work vehicle **100** may be a work machine having work equipment, such as a hydraulic excavator, a wheel roller, a motor grader, and an undercarriage.

[0094] In addition, a part or all of a program executed by a computer in the embodiments described above can be distributed via a computer-readable recording medium or a communication line.

INDUSTRIAL APPLICABILITY

[0095] According to one aspect described above, the path plan of the work vehicle can be efficiently generated.

REFERENCE SIGNS LIST

TABLE-US-00001 10 . . . work vehicle 20 . . . path plan generation device 22 . . . information storage unit 23 . . . path plan generation unit 50 . . . work vehicle path plan generation system 72 . . . position detection unit 73 . . . path plan storage unit 74 . . . vehicle control unit 100 . . . work vehicle 130 . . . work equipment A . . . construction target area D1 . . . earth D5 . . . earth Dm1, Dm5 . . . three-dimensional model G . . . ground L . . . movement path R . . . travel path S . . . design surface

Claims

1. A work vehicle path plan generation system that generates a path plan in order for a work vehicle including work equipment to perform an excavation work on a ground of a construction target area, the work vehicle path plan generation system comprising: a position detection unit configured to detect a position of the work vehicle; an information storage unit configured to store topographic shape information indicating a shape of a topography in the construction target area, a position of the work vehicle, and design surface information indicating a target shape in the construction target area; and a path plan generation unit configured to generate a work equipment path plan indicating a movement path of the work equipment and a travel path plan indicating a travel path of the work vehicle, based on the topographic shape information, the position of the work vehicle, and the design surface information.

2. The work vehicle path plan generation system according to claim 1, wherein the path plan generation unit is subjected to reinforcement learning such that an optimal travel path and an optimal movement path are generated.

3. The work vehicle path plan generation system according to claim 2, wherein a plurality of times of simulations are performed by using at least an earth excavated by the work equipment and a parameter related to the work vehicle on the travel path to perform the reinforcement learning on

the path plan generation unit.

4. The work vehicle path plan generation system according to claim 2, wherein the path plan generation unit is subjected to the reinforcement learning based on a reward obtained by using at least one of an excavation earth amount and a work time in a case where the work vehicle is caused to move along the travel path.
 5. The work vehicle path plan generation system according to claim 4, wherein the path plan generation unit is subjected to the reinforcement learning using the excavation earth amount calculated based on a holding earth amount with the work equipment, which is based on a difference in a topography before and after the work vehicle is caused to move along the travel path to excavate the ground with the work equipment.
 6. The work vehicle path plan generation system according to claim 5, wherein the path plan generation unit is subjected to the reinforcement learning using the holding earth amount calculated by causing the work vehicle to move along the travel path and then modeling an earth held in front of the work equipment as a polygonal prism-shaped three-dimensional model.
 7. The work vehicle path plan generation system according to claim 5, wherein the path plan generation unit is subjected to the reinforcement learning using the excavation earth amount calculated based on an earth amount of wavy windrows protruding on both sides of the work equipment in a case where the work vehicle is caused to move along the travel path.
 8. The work vehicle path plan generation system according to claim 7, wherein the path plan generation unit is subjected to the reinforcement learning using the earth amount of the windrows calculated by modeling an earth of the windrows as a polygonal prism-shaped three-dimensional model.
 9. A work vehicle path plan generation method that generates a path plan in order for a work vehicle including work equipment to perform an excavation work on a ground of a construction target area, the work vehicle path plan generation method comprising: a step of detecting a position of the work vehicle; a step of storing topographic shape information indicating a shape of a topography in the construction target area, a position of the work vehicle, and design surface information indicating a shape required to be excavated of the ground in the construction target area; and a step of generating a work equipment path plan indicating a movement path of the work equipment and a travel path plan indicating a travel path of the work vehicle, based on the topographic shape information, the position of the work vehicle, and the design surface information.
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